Functions, Modulus and Quadratics

ANSWER KEY

Section 1: Function Mapping, Domain/Range and Inverses

1 The functions f and g are defined for $x \in \mathbf{R}$ by $f(x) = 2x - 5$ and $g(x) = x^2 + 3$. Find the composite value $(g \circ f)(4)$.

a. 12 ✓

b. 21

c. 14

d. 9

Evaluate $f(4) = 2(4) - 5 = 3$. Then $(g \circ f)(4) = g(3) = 3^2 + 3 = 12$.

2 A function is defined by $f(x) = \frac{3x+1}{x-2}$ for $x \neq 2$. Find an expression for the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$ ✓

b. $f^{-1}(x) = \frac{x-2}{3x+1}, \quad x \neq -\frac{1}{3}$

c. $f^{-1}(x) = \frac{2x-1}{x+3}, \quad x \neq -3$

d. $f^{-1}(x) = \frac{3x-2}{x-1}, \quad x \neq 1$

Let $y = \frac{3x+1}{x-2} \Rightarrow yx - 2y = 3x + 1 \Rightarrow x(y-3) = 2y + 1 \Rightarrow x = \frac{2y+1}{y-3}$. Thus $f^{-1}(x) = \frac{2x+1}{x-3}$ for $x \neq 3$.

3 The square root function $h(x) = 4 - \sqrt{2x - 6}$ is defined on its maximal real domain. Determine the domain and range of h.

a. **Domain:** $x \geq 3$, **Range:** $h(x) \leq 4$ ✓

b. Domain: $x \geq 3$, Range: $h(x) \geq 4$

c. Domain: $x > 3$, Range: $h(x) \in \mathbf{R}$

d. Domain: $x \leq 3$, Range: $h(x) \leq 0$

For the square root to exist: $2x - 6 \geq 0 \Rightarrow x \geq 3$. Since $\sqrt{2x-6} \geq 0$, $h(x) = 4 - \sqrt{2x-6} \leq 4$.

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Section 2: Modulus Equations and Inequalities

4 Solve the linear modulus equation $|3x - 4| = 2x + 1$ for all valid real solutions.

a. $x = 5$ **and** $x = \frac{3}{5}$ ✓

b. $x = 5$ only

c. $x = -1$ and $x = 3$

d. No real solutions

Case 1: $3x - 4 = 2x + 1 \Rightarrow x = 5$ (valid). Case 2: $3x - 4 = -(2x + 1) \Rightarrow 5x = 3 \Rightarrow x = 3/5$ (valid).

5 Find the complete solution set for the absolute value inequality $|2x - 7| < 5$.

a. $1 < x < 6$ ✓

b. $x < 1$ or $x > 6$

c. $-6 < x < 1$

d. $2 < x < 5$

$-5 < 2x - 7 < 5 \Rightarrow 2 < 2x < 12 \Rightarrow 1 < x < 6$.

6 Solve the two-modulus equation $|x + 2| = |2x - 5|$.

a. $x = 7$ **and** $x = 1$ ✓

b. $x = 7$ and $x = -1$

c. $x = 3$ and $x = -3$

d. $x = \frac{7}{3}$ only

$(x + 2)^2 = (2x - 5)^2 \Rightarrow x^2 + 4x + 4 = 4x^2 - 20x + 25 \Rightarrow 3x^2 - 24x + 21 = 0 \Rightarrow x = 7$ or $x = 1$.

Section 3: Quadratic Discriminant and Line-Parabola Intersections

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7 Find the set of values of k for which the quadratic equation $x^2 + (k+2)x + (2k+1) = 0$ has two distinct real roots.

a. $k < 0$ **or** $k > 4$ ✓

b. $0 < k < 4$

c. $k \leq 0$ or $k \geq 4$

d. $-4 < k < 0$

$\Delta = (k+2)^2 - 4(1)(2k+1) = k^2 - 4k = k(k-4) > 0 \Rightarrow k < 0$ or $k > 4$.

8 The line $y = 2x + c$ is tangent to the parabola $y = x^2 - 4x + 7$. Determine the exact value of the constant c.

a. $c = -2$ ✓

b. $c = 2$

c. $c = -7$

d. $c = 3$

$x^2 - 4x + 7 = 2x + c \Rightarrow x^2 - 6x + (7 - c) = 0$. Tangency requires $\Delta = 36 - 4(7 - c) = 8 + 4c = 0 \Rightarrow c = -2$.

9 Find the range of values of m for which the line $y = mx - 1$ does not intersect the curve $y = 2x^2 + 3x + 1$.

a. $-1 < m < 7$ ✓

b. $m < -1$ or $m > 7$

c. $-7 < m < 1$

d. $m < 3$ or $m > 5$

$2x^2 + (3 - m)x + 2 = 0$. $\Delta = (3 - m)^2 - 16 = (m + 1)(m - 7) < 0 \Rightarrow -1 < m < 7$.

Section 4: Completing the Square and Applied Optimization

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10 By completing the square, express $f(x) = 2x^2 - 12x + 23$ in the form $a(x-h)^2 + k$, and state the minimum value of $f(x)$.

a. $2(x-3)^2 + 5$**, minimum value** 5 ✓

b. $2(x-3)^2 + 23$, minimum value 23

c. $2(x-6)^2 - 13$, minimum value -13

d. $(2x-6)^2 + 5$, minimum value 5

$2(x^2 - 6x) + 23 = 2[(x-3)^2 - 9] + 23 = 2(x-3)^2 + 5$. Minimum value is 5 at $x = 3$.

11 A rectangular field has three sides fenced with 80 m of fencing against a river. Find the maximum possible area.

a. $800\ \text{m}^2$ ✓

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

$A(x) = x(80 - 2x) = -2x^2 + 80x$. Vertex $x = 20$ m. Area $= 20(40) = 800\ \text{m}^2$.

12 Find all real solutions to the biquadratic equation $x^4 - 10x^2 + 9 = 0$.

a. $x = \pm 1$ **and** $x = \pm 3$ ✓

b. $x = 1$ and $x = 9$

c. $x = \pm 1$ only

d. $x = \pm\sqrt{10}$

Let $u = x^2$: $(u-1)(u-9) = 0 \Rightarrow u = 1$ or $u = 9 \Rightarrow x = \pm 1,\ \pm 3$.

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